

ECON2043 Experimental and Behavioural Economics

Seminar 2: Problem Set (Time Preferences)

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General Ideas

1. Rational benchmark

- ▶ Time-consistent (w/o β or $\beta = 1$).

2. Present-biased

- ▶ $\beta < 1$.
- ▶ Immediate gratification & delayed costs.

3. Naive agents

- ▶ Use backward induction, based on the rational benchmark.

4. Sophisticated agents

- ▶ Also use backward induction, but based on future selves' actual choices.

Warm-up

Suppose your preferences over cookies today, C_1 , and cookies tomorrow, C_2 , are captured by

$$U(C_1, C_2) = u(C_1) + \beta u(C_2)$$

In this exercise, we look at the linear case where $u(C_t) = C_t$, where $t \in \{1, 2\}$.

(a) Imagine you are perfectly patient, $\beta = 1$, and that you can choose between having 1 cookie today or 2 cookies tomorrow. What would you choose? What would you choose if, instead, you had to choose between 2 cookies today and 1 cookie tomorrow?

(b) Repeat part (a) but now assume that $\beta = \frac{2}{3}$ or $\beta = \frac{1}{3}$.

(c) Suppose you could eat 1 cookie today or give it to your friend, who will then give you x cookies tomorrow. What is the smallest x that makes you want to give the cookie to your friend when β is $1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}$?

Warm-up Answer

(a)

$$U(1, 0) = 1, \quad U(0, 2) = 2$$

$$U(2, 0) = 2, \quad U(0, 1) = 1$$

Choose 2 cookies in both comparisons.

(b)

$$U(1, 0) = 1, \quad U(0, 2) = 2\beta$$

If $\beta = \frac{2}{3}$, then $U(0, 2) = \frac{4}{3} > 1$, so wait. If $\beta = \frac{1}{3}$, then $U(0, 2) = \frac{2}{3} < 1$, so eat today.

For “2 today vs. 1 tomorrow”, immediate 2 still wins for both values of β .

(c)

$$U(1, 0) = 1, \quad U(0, x) = \beta x$$

Set $1 = \beta x$, so $x = \frac{1}{\beta}$.

$$\beta = 1 \Rightarrow x = 1, \quad \beta = \frac{3}{4} \Rightarrow x = \frac{4}{3}, \quad \beta = \frac{1}{2} \Rightarrow x = 2, \quad \beta = \frac{1}{4} \Rightarrow x = 4$$

Question 2

Solve example 2 from the lectures where Kevin needs to clean his room. The cost of cleaning is $c_0 = 3$, $c_1 = 5$, $c_2 = 8$ for the three respective time periods. Solve for the naive equilibrium when $\beta = \frac{1}{2}$, $\delta = 1$. Compare your answer with that of the sophisticated DM, which was $s^* = (Y, N, Y)$. Who is better off?

Answer 2

The time-consistent strategy is (Y, Y, Y) , so the naive believes he will clean at the next best opportunity.

At $t = 1$:

$$U_1(Y, \hat{a}_2 = Y) = -5, \quad U_1(N, \hat{a}_2 = Y) = \beta(-8) = -4$$

Hence $a_1^* = N$.

At $t = 0$, under the wrong belief:

$$U_0(Y, \hat{s}_{-0} = (Y, Y)) = -3, \quad U_0(N, \hat{s}_{-0} = (Y, Y)) = \beta(-5) = -2.5$$

Hence $a_0^* = N$.

Therefore

$$s^* = (N, N, Y), \quad \hat{s}^* = (N, Y, Y)^1$$

The naive waits until the last period and is worse off than the sophisticated DM, who cleans at $t = 0$.

¹ s^* denotes the actual strategy. $\hat{s}^*$ denotes the naive DM's belief about his own behavior from today's perspective. After choosing N today, he still believes that his future selves will follow the time-consistent benchmark and choose (Y, Y) . Hence, the believed strategy is $\hat{s}^* = (N, Y, Y)$.

Question 3

Example 2 in our lecture was a simplified version (of example 1) from O'Donoghue and Rabin (1999). I made two simplifications. First, I reduced the problem to three periods only. Second, I omitted a fixed reward $\bar{v}$ the DM gets for doing the activity. The reward is independent of when the task is done and is received at some further point in time in the future, i.e., $t \geq 3$ (typically at $T + 1$). Does it matter if we include $\bar{v}$ or not?

Answer 3

If the task gets completed at some point, then including $\bar{v}$ does not change the ranking of options. For example²:

$$U_0(Y, s_{-0} = (N, Y)) = -3 + \beta\delta^\tau \bar{v}$$

$$U_0(N, s_{-0} = (N, Y)) = \beta\delta^2(-8) + \beta\delta^\tau \bar{v}$$

The term $\beta\delta^\tau \bar{v}$ appears in both expressions, so it cancels.

Conclusion: if completion is guaranteed, $\bar{v}$ does not matter. It matters only if there is a chance the task is never completed.

²The continuation strategy (N, Y) is just one example. Using (Y, Y) instead would give $U_0(Y, s_{-0} = (Y, Y)) = -3 + \beta\delta^\tau \bar{v}$ and $U_0(N, s_{-0} = (Y, Y)) = \beta\delta(-5) + \beta\delta^\tau \bar{v}$. The common $\beta\delta^\tau \bar{v}$ term still cancels.

Question 4(a)

A decision maker has to do a task once. He can do it in one of three periods ($t \in \{0, 1, 2\}$).

The cost of doing so for each period is $c_0 = 1$, $c_1 = 8$, $c_2 = 0$ but these are only experienced at $t = 3$. The reward of doing so is experienced immediately and is $v_0 = 0$, $v_1 = 5$, $v_2 = 1$. Let $\delta = 1$ and $\beta = \frac{1}{2}$. What would a sophisticated do? Comment on anything noteworthy in your answer.

Answer 4(a)

At $t = 1$:

$$U_1(Y, a_2 = Y) = 5 + \beta(-8) = 1, \quad U_1(N, a_2 = Y) = \beta(1) = 0.5$$

So the $t = 1$ self chooses Y .

At $t = 0$:

$$U_0(Y, s_{-0} = (Y, Y)) = \beta(-1) = -0.5$$

$$U_0(N, s_{-0} = (Y, Y)) = \beta(5) + \beta(-8) = 2.5 - 4 = -1.5$$

So the $t = 0$ self also chooses Y .

Hence

$$s^* = (Y, Y, Y)$$

Noteworthy: the sophisticated DM chooses $t = 0$ even though $t = 2$ dominates $t = 0$ in direct payoff terms. The reason is strategic fear of the $t = 1$ self.

Question 4(b)

Next, swap the costs and benefits around and assume costs are experienced immediately instead, i.e., $c_0 = 0$, $c_1 = 5$, $c_2 = 1$ and $v_0 = 1$, $v_1 = 8$, $v_2 = 0$. Solve for the naif's equilibrium strategy. Comment on anything noteworthy in your answer.

Answer 4(b)

The time-consistent benchmark is:

$$U_0((Y, \cdot, \cdot)) = 1, \quad U_0((N, Y, \cdot)) = -5 + 8 = 3, \quad U_0((N, N, Y)) = -1$$

So the naive believes the $t = 1$ self will do the task. i.e., (N, Y, Y) .

Actual choice at $t = 1$:

$$U_1(Y, \hat{a}_2 = Y) = -5 + \beta(8) = -1, \quad U_1(N, \hat{a}_2 = Y) = \beta(-1) = -0.5$$

So the $t = 1$ self waits.

At $t = 0$:

$$U_0(Y, \hat{s}_{-0} = (Y, Y)) = \beta(1) = 0.5, \quad U_0(N, \hat{s}_{-0} = (Y, Y)) = \beta(-5) + \beta(8) = 1.5$$

Hence

$$s^* = (N, N, Y), \quad \hat{s}^* = (N, Y, Y)^3$$

~~Again, the naive overestimates the future self and ends up with a dominated outcome.~~

³Here, $\hat{s}^*$ again denotes the naive DM's believed strategy from today's perspective. In this example, it happens to coincide with the time-consistent benchmark, because after choosing N today, the naive still believes that future selves will choose (Y, Y) .

Question 5

Let's go back to example 2 from the lectures, where Kevin needs to clean his room. The cost of cleaning is $c_0 = 3$, $c_1 = 5$, $c_2 = 8$ for the three respective time periods. Recall that a sophisticated DM's equilibrium strategy for $\beta = \frac{1}{2}$, $\delta = 1$ is (Y, N, Y) . Come up with a commitment mechanism that allows a sophisticated Kevin to clean in time $t = 1$ and show (by calculating the payoffs) that it gives rise to the equilibrium of (N, Y, Y) .

Answer 5

One commitment device: Kevin loses pocket money worth x if he has not cleaned by $t = 1$.

At $t = 1$:

$$U_1(Y) = -5, \quad U_1(N) = -x + \beta(-8) = -x - 4$$

To make cleaning at $t = 1$ optimal, we need

$$-5 > -x - 4 \quad \Leftrightarrow \quad x > 1$$

If Kevin knows the $t = 1$ self will clean, then at $t = 0$:

$$U_0(Y, s_{-0} = (Y, Y)) = -3, \quad U_0(N, s_{-0} = (Y, Y)) = \beta(-5) = -2.5$$

So he waits at $t = 0$.

Hence the commitment device gives

$$(N, Y, Y)$$

Question 6

Further practice opportunities: example 1 (p. 109), example 2 (p. 110), and example 3 (p. 112) in O'Donoghue and Rabin (1999). When doing so, note the notational difference to ours: the first period in that paper is $t = 1$ while ours is $t = 0$. More importantly, be careful with regards to the terms “costs (reward) are immediate” and “costs (reward) are delayed”. Costs or rewards from the latter expression accrue in period $T + 1$.

Answer 6

There is no single numerical answer here. The main points to watch are:

- ▶ Convert their timeline correctly: their first period is $t = 1$, ours is $t = 0$.
- ▶ Track when utility is actually experienced.
- ▶ “Immediate” means the payoff or cost is felt in the current period.
- ▶ “Delayed” means it is realized in period $T + 1$.

Presentation for Next Session

Task

- ▶ Form a group of around 4 people with classmates in your seminar (for this task only)⁴.
- ▶ Find 1 paper for the presentation⁵.

Basis

- ▶ Research question, method, experimental design and the main findings, etc.

Further

- ▶ Evaluate whether the idea is interesting, important, novel, and non-obvious.

⁴We know this may make some of you feel a bit unhappy. It could be super cool or a disaster. Why not just treat this as an experiment? Everything can be treated as an experiment, in an experimental module :)

⁵Keep it brief. Two slides are enough, with 1-2 presenters and up to 5 minutes. Please send your paper and slides before the next session.